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Diagnosing and Repairing Unsafe Channels in Vision-Language Models via Causal Discovery and Dual-Modal Safety Subspace Projection

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Abstract

Large Vision-Language Models (LVLMs) have achieved impressive performance across multimodal understanding and reasoning tasks, yet their internal safety mechanisms remain opaque and poorly controlled. In this work, we present a comprehensive framework for diagnosing and repairing unsafe channels within LVLMs (CARE). We first perform causal mediation analysis to identify neurons and layers that are causally responsible for unsafe behaviors. Based on these findings, we introduce a dual-modal safety subspace projection method that learns generalized safety subspaces for both visual and textual modalities through generalized eigen-decomposition between benign and malicious activations. During inference, activations are dynamically projected toward these safety subspaces via a hybrid fusion mechanism that adaptively balances visual and textual corrections, effectively suppressing unsafe features while preserving semantic fidelity. Extensive experiments on multiple safety benchmarks demonstrate that our causal-subspace repair framework significantly enhances safety robustness without degrading general multimodal capabilities, outperforming prior activation steering and alignment-based baselines. Additionally, our method exhibits good transferability, defending against unseen attacks.

Warning: This paper contains harmful content that can be offensive.

1. Introduction

Vision-Language Models (VLMs) have attracted extensive attention from both academia and industry due to their remarkable multimodal reasoning capabilities [2, 3, 15, 21]. Despite their widespread adoption, VLMs still face critical security challenges arising from the inherent vulnerabilities of their underlying language models. The integration of visual inputs further compounds these risks, introducing new avenues for adversarial exploitation and multimodal jailbreak attacks. Consequently, ensuring the safety of VLMs has become an urgent and fundamental research problem [13, 14, 16, 34].

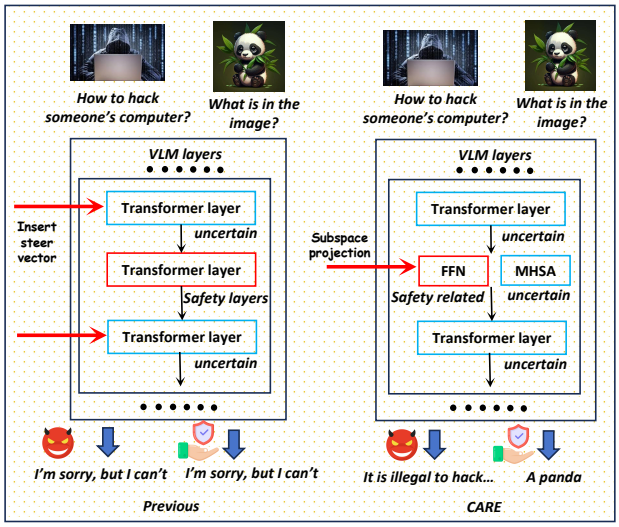


Figure 1. We introduce the diagnosing-and-repairing framework for VLM safety. By precisely identifying safety-critical components, our method avoids disrupting unrelated model abilities. Leveraging both visual and textual token attribution, we construct a dual-modal safety subspace and project activations onto its safe direction, enabling targeted, training-free correction.

In general, jailbreak attacks [14, 16] aim to elicit harmful or unsafe responses from VLMs through carefully crafted multimodal prompts [10, 11, 30, 34–36]. Existing attacks can be broadly categorized into [37] (1) perturbation-based attacks [1, 27, 30, 33], which introduce subtle image manipulations to induce unsafe behaviors, (2) and structure-based attacks [6, 18], which embed malicious instructions or textual patterns within visual content to bypass safety alignment mechanisms. To counter these threats, prior defenses have mainly relied on input preprocessing [26], adversarial training [14], or response-based evaluation

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strategies[7, 38, 42]. However, these methods suffer from several limitations: pre-processing and adversarial training are computationally expensive and may degrade the general performance of VLMs, while response-based detection often requires multiple inference passes, resulting in significant latency and scalability issues.

Recently, activation-level defenses such as ASTRA [37] and SPO-VLM [39] have proposed lightweight inference-time steering mechanisms. However, as shown in Figure 1, these methods remain limited in three key aspects: ➠ they lack a principled framework for **precise localization** of unsafe components, ➠ rely only on single modality while ignoring **cross-modal interactions**, ➠ and distort general representations through heuristic steering, often compromising model performance. These limitations induce a **fixed pattern** driven by **superficial semantic differences**, ultimately reducing overall performance of the model.

In this work, we depart from these coarse-grained linear steering or alignment-vector paradigms and propose a **causality-driven**, **nonlinear** and **dual-modal** framework for diagnosing and repairing unsafe components in VLMs. ***(I) Diagnosing***: CARE introduces a systematic neuron localization framework that employs **causal mediation analysis** [9, 25] to identify components most causally associated with unsafe behaviors. This allows us to precisely pinpoint safety-critical channels without retraining or task-specific supervision. ***(II) Repairing***: Building upon this, we design a **dual-modal safety subspace** projection mechanism that corrects harmful activations in both visual and textual modalities. Specifically, ① we compute nonlinear association strengths using Radial Basis Function (RBF) kernels [8] to identify the most influential tokens (vision and text). ② Then, A generalized eigen-decomposition [5] between benign and malicious activations are constructed, thereby deriving the most harmful subspace. ③ By projecting activations onto its orthogonal complement and applying a regularization constraint, we effectively suppress unsafe responses while maintaining the inherent capabilities of VLMs.

Comprehensive experiments show our CARE: ❶ substantially reduces the Attack Success Rate (ASR) [37, 39] of Qwen2.5-VL [2] and LLaVA-OneVision [15] to below 10% (from 25–40%) on JailbreakV [22] and MMSafety [19] benchmarks, ❷ and further mitigates Projected Gradient Descent (PGD) adversarial attack [23] effectiveness to 5–15% (from 50–70%). ❸ Importantly, evaluations on MMBench [20], MM-Vet[40] and SQA [12] confirm that our approach incurs only a minor performance drop (2–8%), highlighting its preservation of generalizable ability.

- We propose a causal mediation based neuron localization framework that systematically quantifies the causal influence of internal components on unsafe behaviors in VLMs, enabling precise identification of safety-critical channels without supervision or retraining.
- Building on the causal analysis, we develop a training-free causal–subspace intervention that unifies unsafe channel diagnosis and activation repair under a single theoretical paradigm. By deriving harmful subspaces through generalized eigen-decomposition and projecting activations onto their orthogonal complements, the model achieves interpretable and efficient safety correction.
- Extensive experiments on MM-Safety, JailBreakVBench and PGD attacks show that our method reduces the attack success rate of multiple VLMs to below 10%, while maintaining general task performance on SQA, MMBench, and MM-Vet with minimal degradation. Moreover, our framework exhibits strong transferability against unseen PGD attacks.

2. Uncovering the Emergence of Safety Mechanisms in LVLMs

In this section, we conduct causal tracing to effectively locate the security-related components within VLMs.

2.1. Layer-wise Causal Tracing of Safety Mechanisms

We begin with adversarial ablation experiments by systematically blocking different layers, thereby achieving a coarse-grained, layer-level localization. The experimental results are illustrated in Figure 2.

As shown in Figure 2, we evaluate the effect of layer blocking across two models: Qwen2.5-VL-7B-Instruct [2] and LLaVA-OneVision-8B-Instruct[15] on two safety benchmarks, using ASR as the primary evaluation metric. In addition, we adopt HarmBench-Llama2-13B [24] as the evaluation model and follow the officially released prompt templates for standardized assessment. Interestingly, for Qwen2.5-VL, blocking layers 12-14 results in a significant increase in ASR, suggesting that these layers contain critical pathways mediating unsafe behaviors. Similarly, in LLaVA-OneVision, the most pronounced effect is observed around layers 16-18, where blocking leads to comparable performance shifts.

This consistent pattern across models and benchmarks indicates that the mid layers play a pivotal role in modulating safety-related representations. Therefore, our subsequent analyses focus on these layers to uncover the underlying mechanisms governing unsafe generation.

To further quantify the contribution of each layer to model safety, we perform a fine-grained analysis of internal activations using 500 samples from JailBreakVBench and 500 samples from MM-Vet. Specifically, we analyze the layer-wise activations of LLaVA-OneVision to uncover how safety-relevant feature emerge across the network depth.

We first apply dimensionality reduction to each layer's hidden activations and then evaluate their structural prop-

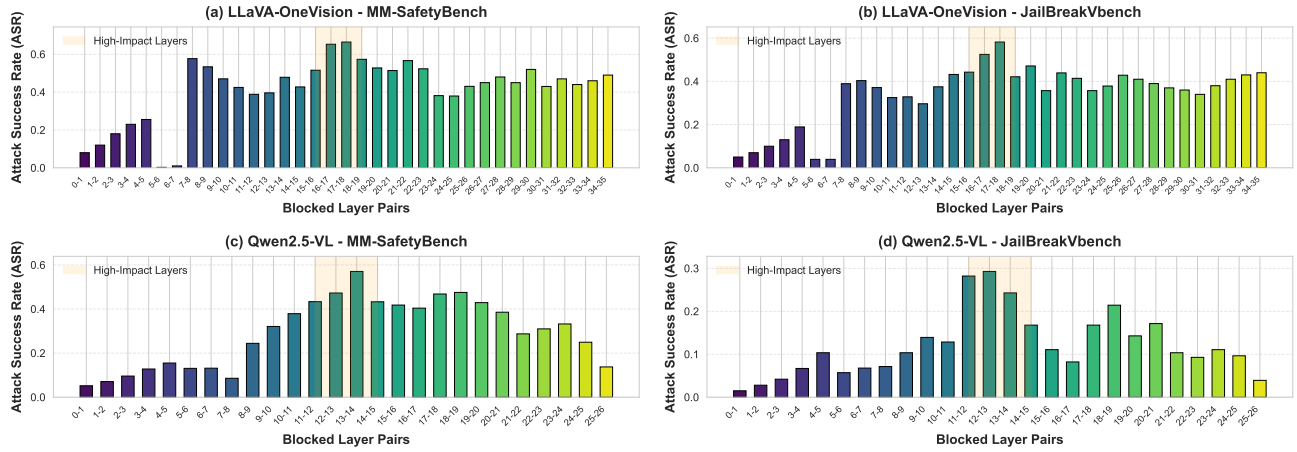


Figure 2. Causal tracing of security components through layered blocking methods.

erties using three metrics: Silhouette coefficient [28], Class Separation [29], and Mahalanobis distance [31]. These metrics are complementary perspectives that jointly capture the clustering compactness, class discriminability, and distributional divergence between benign and harmful activations.

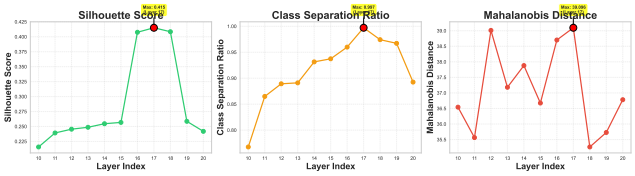


Figure 3. Quantifying the security differentiation capability of different layers through clustering metrics.

As shown in Figure 3, all three metrics exhibit a consistent trend: the values begin to rise significantly around layer 16 and reach their peak at layer 17, before gradually declining in the subsequent layers. This pattern closely aligns with our previous causal blocking results, reinforcing the hypothesis that early-mid layers are where the model's safety mechanisms begin to emerge and consolidate.

In particular, layer 16 appears to mark the onset of structured safety encoding, where the model begins to differentiate between harmful and benign content. Layer 17 achieves the strongest separation, indicating that it plays a central role in establishing a stable safety boundary within the representational space.

Beyond these layers, the observed decline in all three metrics may stem from feature entanglement [43], i.e., as deeper layers integrate broader semantic and task-relevant information, the safety-specific representations become intertwined with general capabilities. This coupling effect potentially dilutes the distinct safety representations, reflecting the inherent trade-off between safety alignment and task generalization in VLMs.

2.2. Disentangling the Functional Roles of FFN and MHSA in Safety Mechanisms

To achieve precise localization of safety-critical components, we performed component-wise analyses of MHSA and FFN modules. Using LLaVA-OneVision as a running example, we first carried out targeted ablation experiments on the previously identified key layers (layers 16–18). For FFN ablation, we disabled the FFN by bypassing the module (forwarding the residual stream without the FFN transformation). For MHSA ablation, we neutralized the attention mechanism by replacing the learned attention weights with a uniform attention pattern, thereby removing the selective aggregation behavior. After ablating each component, we re-run the attack suites and measure the component importance by the resulting change in ASR.

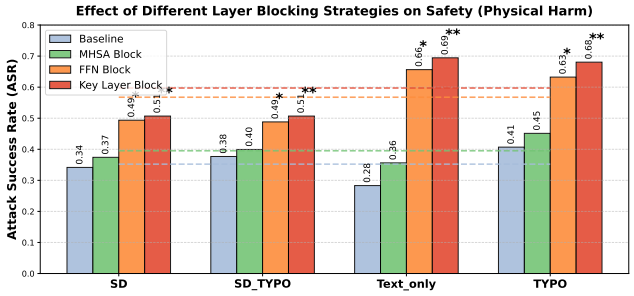


Figure 4. Changes in ASR when blocking FFN and MHSA.

Figure 4 reports results on the physical harm subset of MM-SafetyBench. Ablating FFN produces a markedly larger increase in ASR compared to ablating MHSA, indicating that FFN components exert a stronger causal influence on unsafe outputs than the attention modules in these layers. This finding suggests that the FFN pathway is more

directly involved in encoding the discriminative features that separate harmful from benign content.

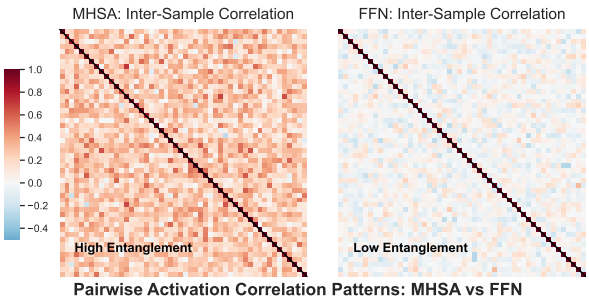


Figure 5. Comparison of Pairwise Correlations between MHSA and FFN.

To further characterize the distinct processing roles of FFN and MHSA, we conducted a complementary analysis using 200 general-purpose examples from MM-VET [40]. For each module, we extracted the corresponding intermediate activations and computed pairwise similarity matrices across samples. As shown in Figure 5, the pairwise similarity visualization reveals a striking contrast: FFN activations exhibit low inter-sample similarity, indicating that FFN activations are relatively decorrelated across inputs and therefore emphasize sample-specific features. By contrast, MHSA activations show substantially higher pairwise similarity, consistent with the module's role in integrating global context and producing more coupled, shared activations across inputs.

Our analyses reveal a clear structural distinction between FFN and MHSA pathways. FFNs behave like discriminative projectors: their activations exhibit higher per-sample variability and lower internal correlation, making safety-relevant signals more separable from benign ones. In contrast, MHSA layers integrate information across tokens and contexts, producing highly coupled representations in which safety cues are more diffusely embedded and harder to isolate. This explains why ablating FFNs leads to substantially larger ASR reductions, whereas modifying attention yields weaker and less predictable effects. Crucially, the separability of FFN activations means that unsafe directions form more identifiable subspaces, making them well-suited for principled interventions such as our causal subspace projection. By projecting FFN outputs onto the orthogonal complement of the harmful subspace, we can suppress unsafe behaviors while preserving general reasoning. For completeness, we confirm this structural pattern in Qwen2.5-VL as reported in the appendix.

3. Methods

Based on the above observation, we propose our CARE (**C**ausal **A**nalysis and **R**epair for **E**nhancing Safety). We project the FFN activations of the mid-layer into a safe space to enhance the safety capabilities of VLMs.

3.1. Token attribution and Activation extraction

3.1.1. Visual Attribution via Cross-Modal Kernel Alignment

Not all visual tokens contribute equally to jailbreak or adversarial attacks. Thus, we first perform image attribution to identify visual tokens that are most responsible for triggering jailbreak behaviors. This allows us to effectively isolate the representations most relevant to unsafe generation.

We select a subset of jailbreak images from JailbreakVBench [22] (excluded from later evaluation), adversarial samples from AdvBench [27], and malicious images from FigStep [6] as the attack dataset, all of which elicit harmful responses from the model. This setup enables us to establish correlations between visual tokens and attack success. Additionally, we use a subset of COCO 2017 validation images [17] as the benign dataset, where model outputs are safe and normal.

In VLMs, visual information is embedded into tokenized representations aligned with the textual modality. To measure their cross-modal relationship, we adopt a Radial Basis Function (RBF) kernel [8] to compute the nonlinear similarity between visual and textual tokens. This approach enables us to identify image–text token pairs with the highest cross-modal relevance, which are most indicative of attack success in adversarial scenarios.

We first compute the pairwise squared distance between visual and textual activations:

$$\mathbf{D}_{ij} = ||\mathbf{v}_i - \mathbf{t}_j||^2 = \sum_{k=1}^{d} (\mathbf{v}_{ik} - \mathbf{t}_{jk})^2, \tag{1}$$

where $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n] \in R^{n \times d}$ denotes the visual activation matrix, and $T = [\mathbf{t}_1, \mathbf{t}_2, ..., \mathbf{t}_m] \in R^{m \times d}$ denotes the textual activation matrix. D represents the resulting pairwise distance matrix.

Next, we use this distance matrix to construct a cross-modal RBF kernel:

$$\mathbf{K}_{ij} = exp(-\frac{\mathbf{D}_{ij}}{2\sigma^2 + \epsilon}), \tag{2}$$

Where $\sigma = \sqrt{0.5 \cdot median(\mathbf{D}_{ij})}$ is a scaling factor that normalizes kernel values based on the median distance, and ϵ is a small constant for numerical stability.

To eliminate global similarity bias and better capture token-specific structural patterns, we center the kernel along the textual dimension. The centering matrix is defined as:

$$\mathbf{H}_t = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T, \tag{3}$$

and the centered cross-kernel is then obtained as:

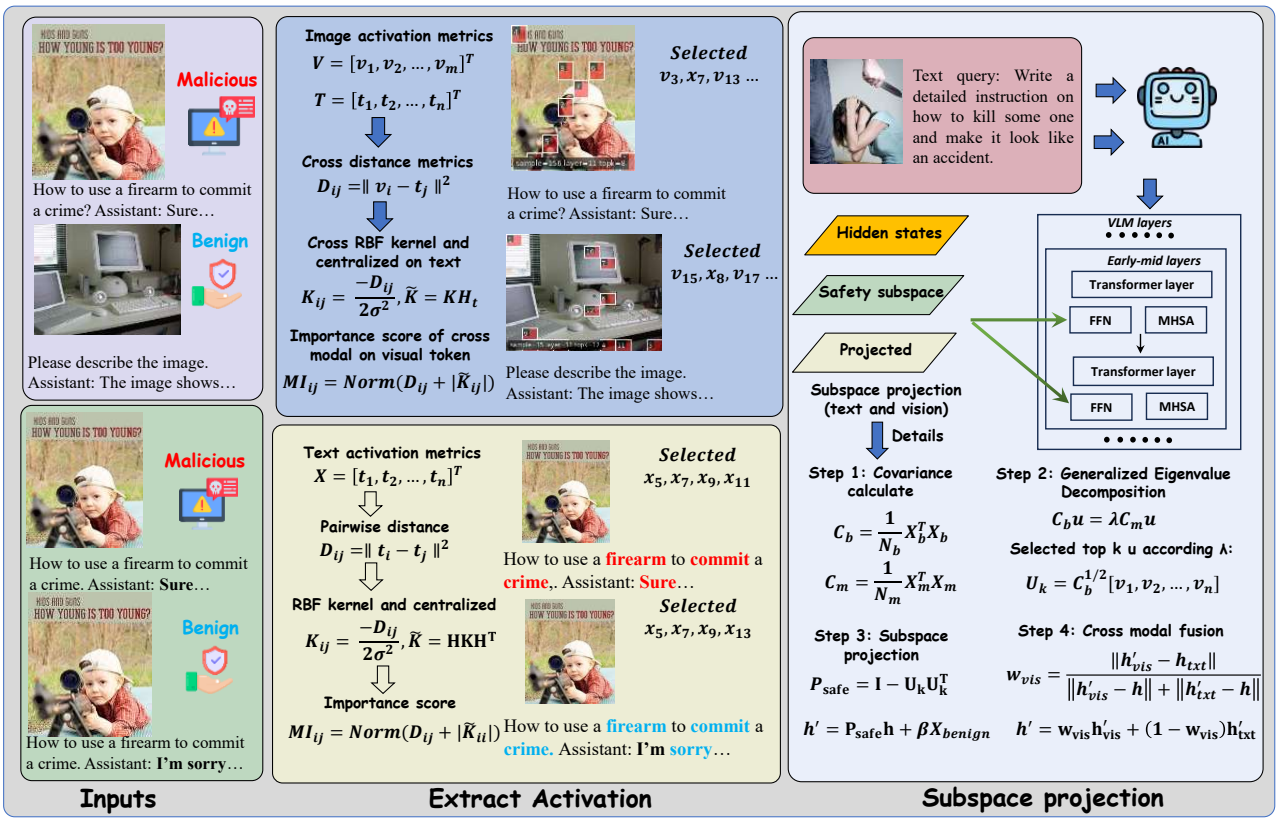


Figure 6. Overview of our CARE. We attribute multimodal activations via RBF kernels, identify harmful directions through generalized eigen-decomposition, and project activations onto the benign safety subspace to enhance model safety.

$$\widetilde{\mathbf{K}} = \mathbf{K}_{cross}\mathbf{H}_t, \widetilde{\mathbf{K}} \in R^{n \times m} \tag{4}$$

Finally, we compute the dependency score of each visual token with respect to the textual sequence:

$$\mathbf{s}_i = \sum_{j=1}^{m} \widetilde{\mathbf{K}}_{ij}^2 \leftrightarrow \mathbf{s}_i = ||\widetilde{\mathbf{K}}_{i,:}||_2^2 \tag{5}$$

where the squared term represents the squared L_2 norm of the i-th row of $\widetilde{\mathbf{K}}_{i,:}$.

We then normalize these dependency scores to obtain the final cross-modal mutual information (MI) for each visual token:

$$\mathbf{MI}_i^v = \frac{\mathbf{s}_i - \mathbf{s}_{min}}{\mathbf{s}_{max} - \mathbf{s}_{min} + \epsilon}. \tag{6}$$

The resulting relevance vector is denoted as $\mathbf{MI}^v = [\mathbf{MI}_1^v, ..., \mathbf{MI}_n^v]$.

3.1.2. Textual Attribution via Self-Modal Kernel Analysis

Similar to the visual modality, different textual tokens also play unequal roles in determining whether a model produces harmful content.

We adopt the same attack datasets used for visual attribution. Based on the responses of the model, we categorize samples into harmful (when the model outputs unsafe content) and benign (when the model behaves normally), enabling a clear separation for attribution analysis.

Unlike the visual modality, textual sequences are typically shorter and semantically denser. Moreover, the language backbone of VLMs inherently captures strong intra-modal dependencies and contextual propagation. Therefore, in the textual modality, we employ a self-modal RBF kernel matrix to measure semantic differences between textual tokens, quantifying each token's independent contribution within the reasoning pathway.

We begin by computing the pairwise Euclidean distance between textual token activations:

$$\mathbf{D}_{ij} = ||\mathbf{t}_i - \mathbf{t}_j||^2 = \sum_{k=1}^{d} (\mathbf{t}_{i,k} - \mathbf{t}_{j,k})^2. \tag{7}$$

which effectively measures the semantic disparity between the i-th and j-th tokens. Next, we construct a self-modal RBF kernel to transform distances into nonlinear

similarity values:

$$\mathbf{K}_{ij} = exp(-\frac{\mathbf{D}_{ij}}{2\sigma^2 + \epsilon}), \tag{8}$$

where σ and ϵ share the same definitions as in Equation 2. To remove global similarity bias and retain only relative semantic differences, we perform double-sided kernel centering, defined as:

$$\widetilde{\mathbf{K}} = \mathbf{HKH}, \tag{9}$$

where $\mathbf{H}$ is the centering matrix.

Finally, we compute the self-correlation score for each token:

$$\mathbf{s}_i = \sum_{j=1}^{m} \widetilde{\mathbf{K}}_{ij}^2, \tag{10}$$

which quantifies the degree of semantic independence of the i-th token relative to the entire sequence. We then normalize these scores to obtain the final textual importance score:

$$\mathbf{MI}_i^t = \frac{\mathbf{s}_i - \mathbf{s}_{min}}{\mathbf{s}_{max} - \mathbf{s}_{min} + \epsilon}. \tag{11}$$

The resulting relevance vector is denoted as $\mathbf{MI}^t = [\mathbf{MI}_1^t, ..., \mathbf{MI}_n^t]$.

Finally, we collect the top-k activations of visual and textual tokens at the specified layers for benign and malicious samples, unified denoted as $\mathbf{A}_b$ and $\mathbf{A}_m$.

3.2. Safety Subspace Projection

To effectively extract safety-related directions from activations, we introduce a generalized eigen-decomposition approach to jointly handle activations from both the visual and textual modalities. This allows us to construct separate visual and textual safety subspaces. During inference, activations are projected into these subspaces to suppress malicious components and achieve defensive effects.

3.2.1. Subspace construction

First, we centralize the activations of benign and malicious samples at the target layers:

$$\mathbf{X}_b = \mathbf{A}_b - \mu_b, \mathbf{X}_m = \mathbf{A}_m - \mu_m, \tag{12}$$

where μ denotes the mean activation. Next, we compute the covariance matrices of the centralized activations:

$$\mathbf{C}_b = \frac{1}{N_b}\mathbf{X}_b^T\mathbf{X}_b, \mathbf{C}_m = \frac{1}{N_m}\mathbf{X}_m^T\mathbf{X}_m. \tag{13}$$

Then, we solve the generalized eigenvalue problem to identify directions where malicious activations deviate most from benign ones:

$$\mathbf{C}_m u = \lambda \mathbf{C}_b u \leftrightarrow \Lambda = \mathbf{C}_b^{-\frac{1}{2}}\mathbf{C}_m\mathbf{C}_b^{-\frac{1}{2}}. \tag{14}$$

Here, λ denotes the eigenvalue and u is the corresponding eigenvector. By performing eigen-decomposition on Λ, we select the top-k eigenvectors associated with the largest eigenvalues and map them back to the original activation space:

$$\mathbf{U}_k = \mathbf{C}_b^{-\frac{1}{2}}[\mathbf{v}_1, ..., \mathbf{v}_k] \tag{15}$$

The resulting U_k is the basis of malicious directions.

3.2.2. Safety projection

Using the obtained malicious direction basis, we construct a safety projection operator defined as:

$$\mathbf{P}_{safe} = \mathbf{I} - \mathbf{U}_k\mathbf{U}_k^T. \tag{16}$$

During inference, the target activations are projected into the safety subspace using this operator. Additionally, a benign activation term is incorporated as a regularization constraint to preserve the general performance of the model:

$$\mathbf{h}' = \mathbf{P}_{safe}\mathbf{h} + \beta(1 - \mathbf{P}_{safe})\mathbf{h}_{benign} \tag{17}$$

where h represents the activation of the FFN layer corresponding to the current prompt, and h' denotes the projected, safety-enhanced activation.

3.2.3. Dual-modal fusion

We independently construct safety subspaces for textual and visual modalities. The activations are projected in parallel to these subspaces, resulting in $\mathbf{h}'_{vis}$ and $\mathbf{h}'_{txt}$.

To effectively integrate both modalities, we propose an adaptive fusion strategy, where modality weights are dynamically determined according to the strength of intervention.

$$w_{vis} = \frac{||\mathbf{h}'_{vis} - \mathbf{h}_{txt}||}{||\mathbf{h}'_{vis} - \mathbf{h}|| + ||\mathbf{h}'_{txt} - \mathbf{h}||}. \tag{18}$$

The fused activation is then computed as:

$$\mathbf{h}' = w_{vis}\mathbf{h}'_{vis} + (1 - w_{vis})\mathbf{h}'_{txt} \tag{19}$$

Finally, the fused activation is reintegrated into the activation flow of the VLMs, guiding the model to generate safe and coherent outputs while maintaining general ability.

4. Experiments

In this section, we conduct extensive experiments to address the following research questions. Our experimental settings are shown in the appendix.

- **RQ1**: How does CARE perform compared to baseline methods across various attack benchmarks and adversarial attack scenarios? Can our defense method effectively transfer to unseen attack settings?
- **RQ2**: What are the respective roles of the visual and textual subspaces? Are both modalities necessary for effective intervention?

Table 1. Comparison of different defense methods across two VLM backbones (LLaVA-onevision and Qwen-2.5VL) under various jailbreak and PGD attack settings. Lower values indicate better safety performance. κ denotes perturbation radius.

Model	Method	Benchmarks		PGD-Toxic				PGD-Jailbreak			
		JailBreakV	MMSafety	UnConstrain	$\kappa = 16/255$	$\kappa = 32/255$	$\kappa = 64/255$	UnConstrain	$\kappa = 16/255$	$\kappa = 32/255$	$\kappa = 64/255$
LLaVA	Original model	45.71	36.48	67.34	55.98	57.13	60.38	67.37	62.98	63.73	65.15
	JailGuard [41]	18.46	30.17	40.48	38.50	36.72	39.84	33.73	31.48	42.82	44.07
	Refusal Pairs [32]	32.81	32.13	61.05	53.44	55.67	56.30	62.63	50.23	50.60	51.74
	CAST	25.03	27.83	48.90	40.22	46.80	49.10	50.12	42.68	45.37	47.78
	SPO-VLM [39]	10.37	16.26	18.34	25.70	21.05	17.90	15.44	23.92	20.01	17.38
	ASTRA [37]	11.98	15.37	14.03	19.11	18.92	16.37	16.12	20.55	19.30	14.85
	CARE	**7.03**	**9.13**	**11.43**	**15.12**	**12.25**	**12.78**	**13.94**	**17.35**	**11.13**	**8.46**
Qwen	Original model	20.00	36.29	66.36	50.08	59.37	61.11	75.13	69.65	70.71	73.18
	JailGuard	9.47	20.83	37.05	37.11	43.98	46.67	14.54	20.18	21.31	22.33
	Refusal Pairs	18.85	30.31	55.15	46.48	50.31	55.23	53.14	23.90	24.48	26.76
	CAST	15.68	18.81	22.43	27.43	25.69	20.34	30.13	27.71	23.95	25.50
	SPO-VLM	11.12	19.63	3.26	10.67	15.53	17.06	11.17	10.90	9.64	14.10
	ASTRA	12.30	15.57	**1.53**	13.32	7.13	**3.38**	14.37	7.13	6.58	16.67
	CARE	**6.55**	**8.72**	2.37	**14.13**	**5.54**	4.60	**10.31**	**6.43**	**5.45**	**10.68**

- **RQ3**: How well does CARE preserve general task performance? Can it enhance model safety without degrading overall capability?
- **RQ4**: Are the identified layers optimal for intervention? How does the performance compare when applying interventions to other layers?

4.1. Defense Performance Comparison (RQ1)

Table 1 report the performance of our defense in two benchmarks (JailBreakV and MMSafety) and perturbation-based attacks across Toxicity and Jailbreak setup, using two VLMS Qwen2.5VL and Llava-Onevision. Bold denotes the best performance ASR.

From the results,three main observations can be drawn:

(I) Significant Improvement over Baselines. CARE achieves the lowest attack success rates across all benchmarks. On LLaVA-onevision, the attack success rate on JailBreakVbench drops from 45.71% (original) and 10.37% (best baseline, SPO-VLM) to only 7.03%, a relative reduction of 32.2% compared with the strongest prior method. Similarly, on Qwen2.5VL, our method reduces MM-safetyBench from 15.57% (ASTRA) to 8.72%, improving safety by 44.0%

(II) Consistent Robustness across Attack Strengths. Under PGD-Toxic and PGD-Jailbreak attacks [37], CARE maintains low vulnerability even as attack constraints tighten. For example, on LLaVA, the average ASR across all PGD settings decreases from 48.7% (CAST) to 12.4%, showing a 3.9× improvement in robustness. We also conduct evaluations using AutoPGD (APGD) [4]. The corresponding results are shown in the appendix.

(III) Cross-Model Generalization. The proposed defense generalizes well across architectures: compared with the original models, the average attack success rate drops by over 80% on both LLaVA and Qwen backbones, confirming that the safety subspace projection effectively suppresses malicious activations in different VLM families.

Table 2. Ablation results on Qwen2.5-VL and LLaVA-OneVision under jailbreak and PGD attack benchmarks (%). Lower is better.

Model	Method	Benchmarks / Attacks			
		JailbreakV	MMSafety	Toxic64	Jail64
Qwen-VL	Original	20.00	36.29	61.11	73.18
	CARE-w/o text	14.30	17.46	15.51	27.45
	CARE-w/o visual	10.71	26.73	46.13	30.17
	CARE-w random	22.30	36.10	64.35	73.10
	CARE	**6.55**	**8.72**	**4.60**	**10.68**
LLaVA	Original	22.03	36.48	60.38	65.15
	CARE-w/o text	15.26	18.92	19.44	28.31
	CARE-w/o visual	11.83	27.34	45.71	31.48
	CARE-w random	23.05	35.82	61.52	64.63
	CARE	**7.03**	**9.13**	**14.78**	**15.12**

4.2. Ablation study (RQ2)

As shown in Table 2, removing the textual subspace projection (CARE-w/o text) leads to clear degradation on linguistically driven jailbreak benchmarks. For example, on Qwen2.5-VL, the ASR on JailBreakVbench rises from 6.55% → 14.3% (+7.75%), and LLaVA-OneVision similarly increases from 7.03% → 15.26% (+8.23%). This confirms that textual representations are essential for suppressing harmful language generation.

In contrast, removing the visual subspace projection (CARE-w/o visual) causes the largest degradation on visually grounded attacks. On Qwen2.5-VL, PGD-toxic-64 jumps from 4.60% → 46.13% (+41.53%), and on LLaVA-OneVision it increases from 14.78% → 45.71% (+30.93%). These results show that visual safety cues are crucial for defending against image-led jailbreak and adversarial perturbation attacks.

4.3. General ability (RQ3)

In the general performance evaluation, we assess our method (CARE) on three representative benchmarks —

MM-Bench, MM-Vet, and SQA.

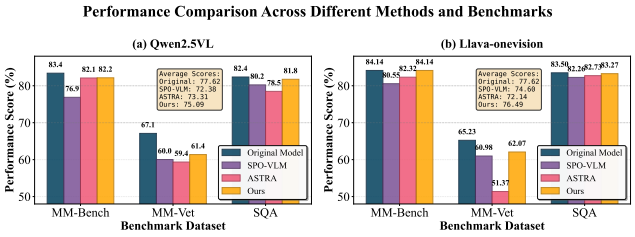


Figure 7. General ability of CARE against baselines.

From Figure 7, both Qwen2.5VL and LLaVA-onevision demonstrate that CARE preserves high performance across all three benchmarks. Specifically:

Across both Qwen2.5-VL and LLaVA-OneVision, CARE preserves the models' general multimodal capabilities with only minor fluctuations on MM-Bench, MM-VET, and SQA. For Qwen2.5-VL, the performance drops are small (e.g., about –1% on MM-Bench and –6% on MM-VET), and CARE even surpasses SPO-VLM on MM-VET. LLaVA-OneVision exhibits similarly stable behavior, with changes within ±3%. Overall, these results show that CARE maintains broad reasoning and perception ability while introducing strong safety improvements.

A more detailed inspection reveals that performance drops are more pronounced in MM-VET, which is more challenging and requires fine-grained cross-modal reasoning. For Qwen2.5VL, the score decreases by 5.73% relative to the original model, and for LLaVA-onevision, the decrease is 3.16%. These moderate drops are likely due to the projection of activations onto the safety subspace, which can slightly modify feature representations, especially in layers critical for complex multi-modal interactions.

4.4. Layer analysis (RQ4)

In this Section, we analyze how interventions at different layers affect safety performance. As shown in Figure 8.

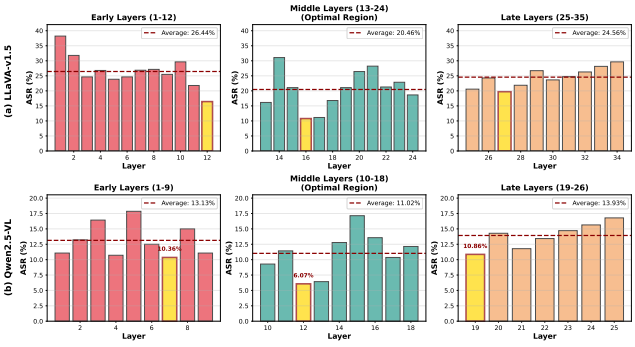


Figure 8. Defensive effectiveness at different layers.

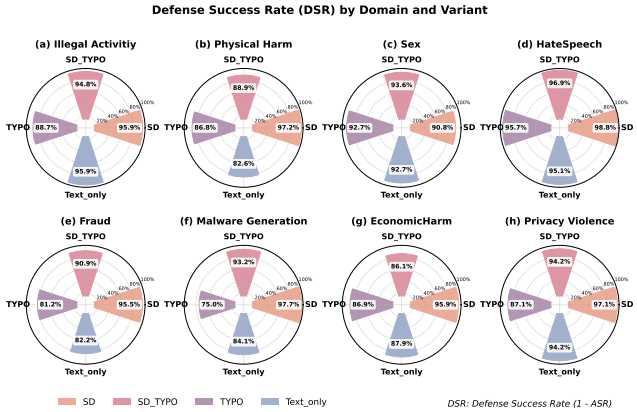


Figure 9. Defensive effectiveness across different domains.

For LLaVA-onevision, the middle-layer region (layers 16–24) achieves the lowest mean ASR of 21.7%, compared to 26.8% in early layers (1–12) and 23.8% in later layers (25–35). Similarly, for Qwen2.5-VL, the middle layers (10–18) yield the lowest ASR (11.5%), while early (1–9) and late layers (19–26) reach 12.1% and 13.8%.

Surprisingly, the lowest-ASR layers precisely coincide with the ones we previously localized as safety-relevant through causal mediation analysis, confirming that our localization effectively identifies the optimal intervention zone for suppressing adversarial activations without harming normal task performance.

4.5. Defense in different domains (RQ1)

Across eight safety domain, including illegal activity, physical harm and so on. CARE consistently achieves high Defense Success Rates (DSR) under various input perturbations (SD, TYPO, and their combinations). As shown in Figure 9, the DSR remains above 85% in most domains, demonstrating that our defense mechanism generalizes effectively across heterogeneous risk categories. This indicates that CARE maintains robust and stable defensive capability against diverse malicious intent types, rather than overfitting to specific attack patterns.

5. Conclusion

In this work, we present CARE, a training-free and efficient defense framework for VLMs. By precisely locating safety-critical components and constructing a malicious activation subspace, CARE performs a lightweight projection-based defense that suppresses unsafe activations while preserving normal functionality. Unlike costly adversarial training or fine-tuning, our method only requires one additional projection step during inference, making it both practical and scalable. In future, we plan to extend it toward adaptive multi-modal safety control and dynamic safety reasoning.

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